

The Transformative Power of GeoAI: A case of study for smallfarmers EUDR compliance

MASTERS DAY 5

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ZGIS



GeoCitizens

 NS

CONTENT

01

MOTIVATION

04

OUTPUTS

02

OBJECTIVES

06

DISCUSSION

How GeoAI-powered digital tools can bridge the gap between EUDR compliance requirements & smallholder farmers' demands

The Transformative Power of GeoAI

From the theoretical & technological foundation

Case Study: Bridging EUDR Compliance for Smallholder Farmers

The human & practical implications of GeoAI

- 01 Establish the relevance of GeoAI for compliance & sustainability workflows
- 02 Design a scalable & replicable GeoAI-based compliance pipeline
- 03 Develop a user-centered dashboard as a tangible solution for the objectives 1 and 2

OUTPUTS

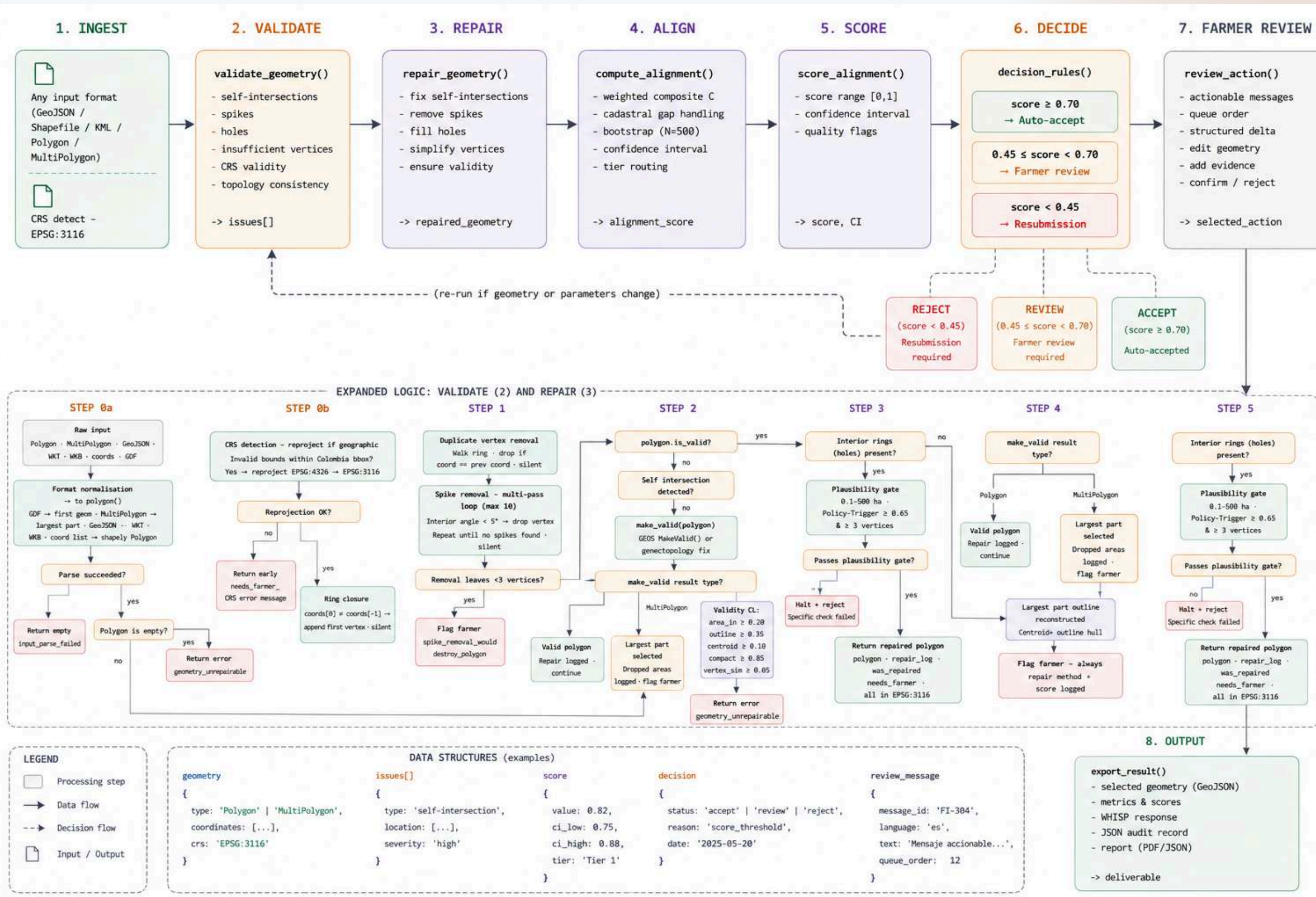
01 A reusable and scalable GeoAI Compliance Framework for broader adoption

02 A dashboard implementing the workflow & centralizing geospatial attributes for smallfarmers



RESULTS

PIPELINE



P0 HARMONIZATION

P1 TOPOLOGY REPAIR

P2 ACTIVE LEARNING FARMER QUEUE

P3 WHISP MODEL INTEGRATION

P4* ADDITIONAL EVIDENCE

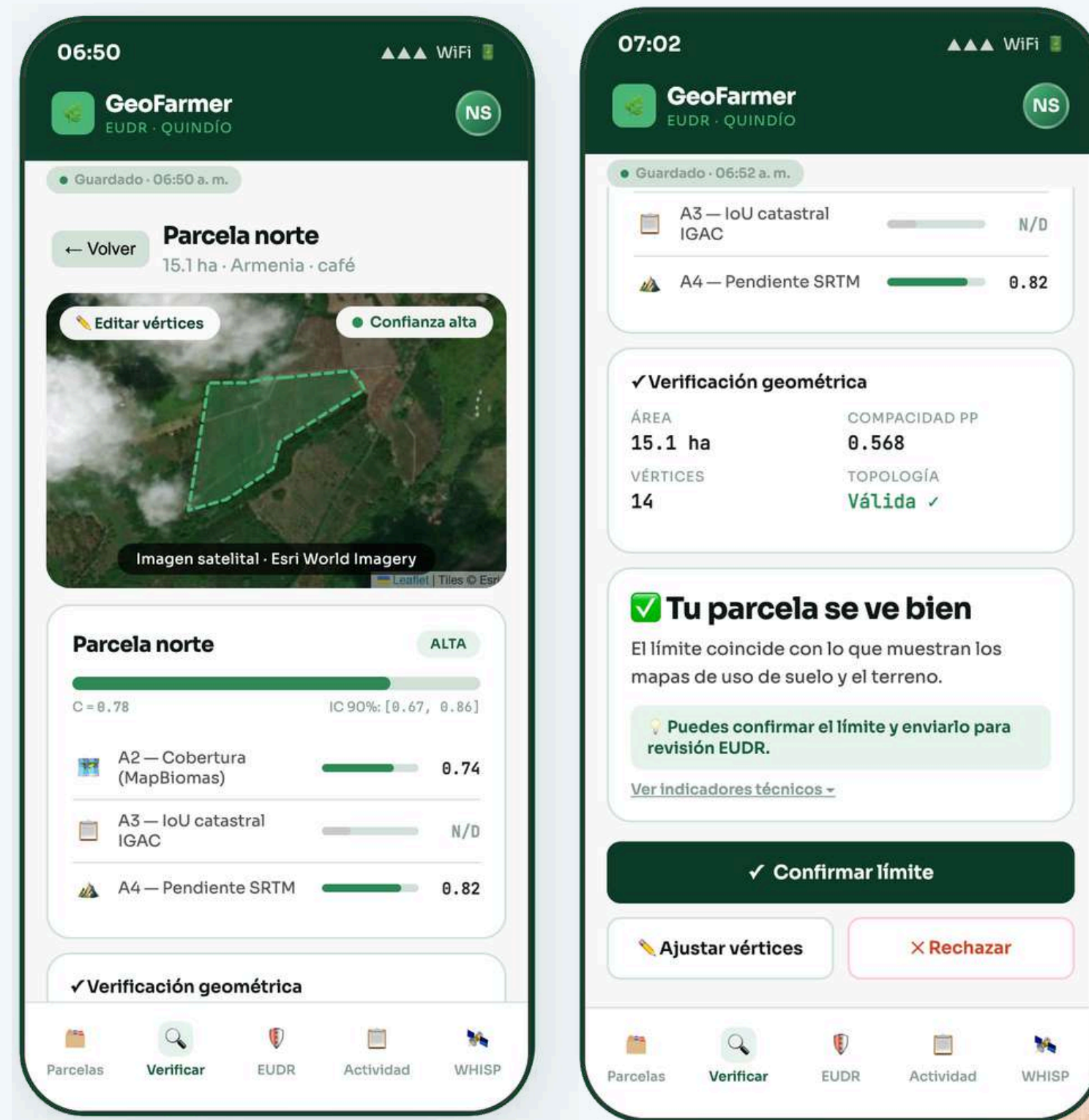


PARCELS WINDOW

- PLOT SUMMARY
- LIST OF PLOTS
- GEOMETRY CONFIDENCE SCORE
- UPLOAD OPTION
- REMOVE ALL OPTION

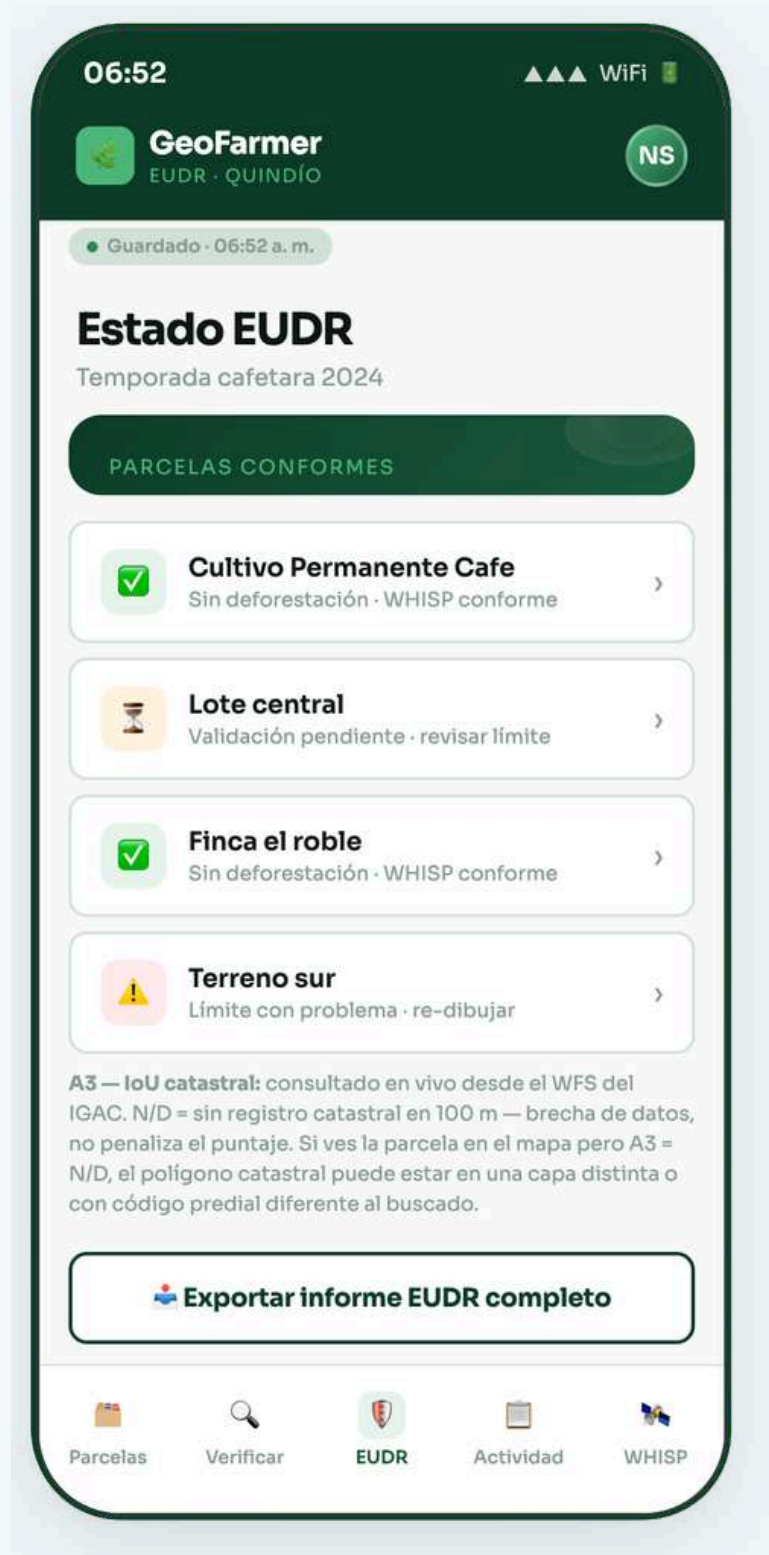
RESULTS

DASHBOARD



VERIFICATION WINDOW

- GEOSPATIAL COMPONENT
- EDIT VERTICES OPTION
- GEOMETRY VALIDATION SUMMARY
- GEOMETRY RESULT DESCRIPTION
- CONFIRM/ADJUST/REJECT OPTIONS



EUDR WINDOW

- CERTIFIED PLOTS SUMMARY
- EUDR ATTRIBUTE TABLE
- PLOT VALIDATION DESCRIPTION
- FULL COMPLIANCE PDF REPORT

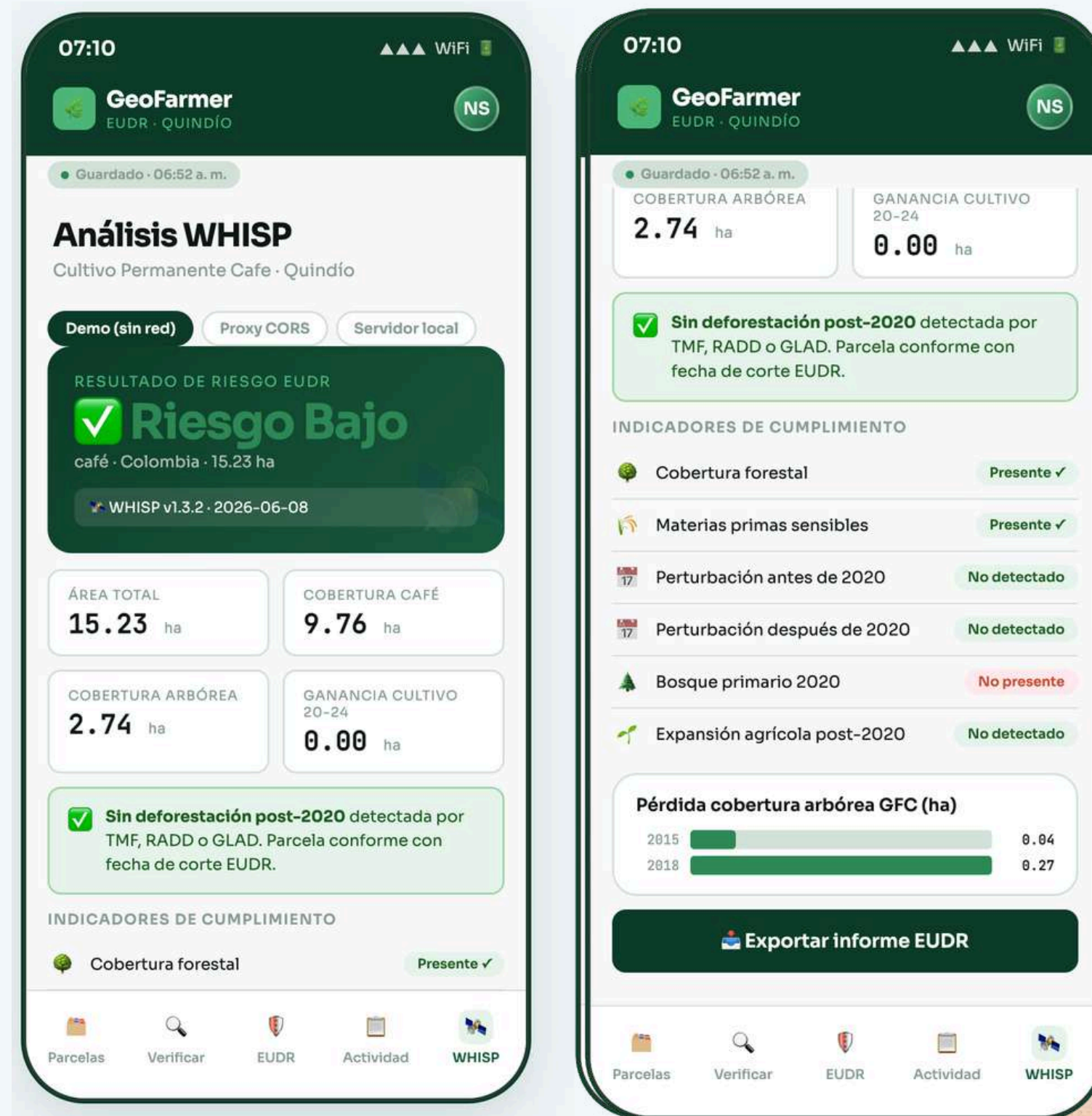


ACTIVITY WINDOW

- EDITS CONFIRMATION
- REJECTIONS LOG
- WHISP SUBMISSIONS
- CONFIDENCE SCORES

RESULTS

DASHBOARD



WHISP WINDOW

- GFC FOREST LOST
- TREE-COVER %
- NDVI VEGETATION OVERLAY
- COMPLIANCE VEREDICT
- EXPLORE FULL COMPLIANCE REPORT OPTION

EUDR REPORT

GeoFarmer — Resumen EUDR 2026-06-08

Reglamento UE 2023/1115 · Fecha de corte: 31 dic 2020 · 4 parcelas

Parcela	Área	C	A2	A3 IGAC	A4	WHISP	GFC post-2020	Estado
Cultivo Permanente Cafe	~12 ha	0.78	0.74	N/D	0.82	low	0.000 ha	Conforme ✓
Lote central	8.4 ha	0.54	0.41	N/D	0.71	no analizado	0.000 ha	Revisar ⚠
Finca el roble	22.7 ha	0.83	0.78	0.79	0.88	no analizado	0.000 ha	Conforme ✓
Terreno sur	5.2 ha	0.31	0.28	N/D	0.38	no analizado	0.000 ha	Revisar ⚠

Generado: 2026-06-08T05:13:37.884Z · GeoFarmer · WHISP OpenForis

COMPLIANCE REPORT

GeoFarmer — Informe de Cumplimiento EUDR

Parcela analizada

✓

CONFORME

Reglamento UE 2023/1115 · Fecha de corte: 31 dic 2020

1. Identificación de la parcela

CAMPO	VALOR
Nombre	Cultivo Permanente Cafe
Área (geométrica)	15.2 ha
Área (WHISP)	15.12 ha
País	Colombia
Departamento	Quindío
Puntuación compuesta (Etapa 2)	0.66 — IC 90%: [0.60, 0.72]
Fecha del informe	2026-06-08
Versión WHISP	1.3.2

2. Veredicto de riesgo WHISP

INDICADOR	RESULTADO
Nivel de riesgo EUDR	Bajo
Pérdida cobertura arbórea post-2020 (GFC)	0.000 ha ✓
Cobertura café (FDaP)	12.84 ha
Cobertura arbórea 2020	8.31 ha
Ganancia agrícola 2020-2024	0.00 ha

3. Indicadores de cumplimiento EUDR

Exportar informe EUDR completo

Parcelas

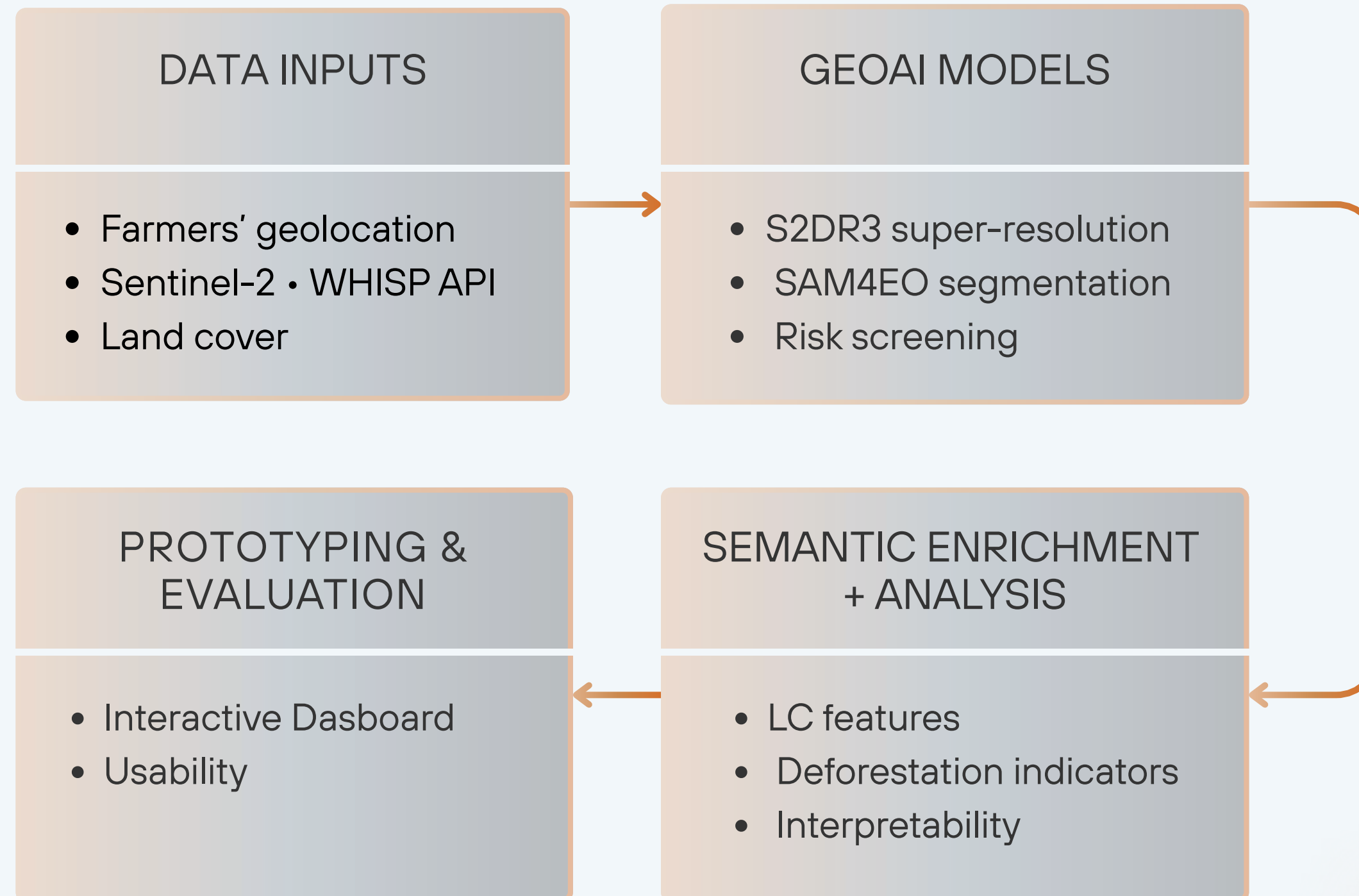
Verificar

EUDR

Actividad

WHISP

– OUTPUTS – PIPELINE'S LAYOUT



— CONCLUSION



Bridge the gap between geospatial intelligence and low-capacity end users under regulatory pressure



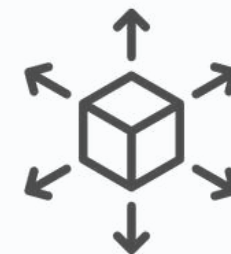
**DIGITAL
INCLUSION**



REGTECH



**GEOSPATIAL
DEMOCRATIZATION**



SCALABILITY



**THIS IS WHAT IT'S
ALL ABOUT**

THANK YOU

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